

The Stochastic Smoluchowski Kramers Approximation with a Mean Field Nonlinearity

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Motivation: The Stochastic Quantization Program

Goal:

Sample from a euclidean quantum field theory described via a target measure

$$d\mu[\phi] = \frac{1}{Z} e^{-S_E[\phi]} d\mathcal{D}\phi.$$

Challenge:

Direct sampling from μ is highly intractable.

Strategy: Stochastic Quantization

Construct stochastic Langevin dynamics ϕ_τ such that:

- 1 **Invariance:** μ is the unique equilibrium state.
- 2 **Convergence:** For an easy-to-sample initial distribution $\phi_0 \sim \rho$,

$$P(\phi_\tau) \xrightarrow{\tau \rightarrow \infty} \mu[\phi].$$

$\Rightarrow$ *Sample from μ by computing ergodic averages of ϕ_τ for large Burn-in times.*

Background: The Hyperbolic $O(N)$ ϕ_2^4 -Model

The target Euclidean QFT on $\mathbb{T}^2$ is reached by sampling the first marginal of:

$$d\rho_N[\phi_N, \pi_N] = \frac{1}{Z_N} \exp \left[-S_E[\phi_N] - \frac{1}{2} \int_{\mathbb{T}^2} \sum_{j=1}^N |\pi_{N,j}|^2 dx \right] \mathcal{D}\phi_N \mathcal{D}\pi_N$$

where $\pi_N = \partial_\tau \phi_N$ is the auxiliary fictitious momentum, and the Euclidean action is:

$$S_E[\phi_N] = \int_{\mathbb{T}^2} \left(\frac{1}{2} \sum_{j=1}^N (|\nabla \phi_{N,j}|^2 + |\phi_{N,j}|^2) + \frac{1}{4N} : \left(\sum_{j=1}^N \phi_{N,j}^2 \right)^2 : \right) dx.$$

Hyperbolic Stochastic Quantization

[Tolomeo, 2023], [Liu, 2025]: ρ_N is the unique equilibrium state of ($1 \leq j \leq N$):

$$\partial_\tau^2 \phi_{N,j} + \partial_\tau \phi_{N,j} + (1 - \Delta) \phi_{N,j} + \frac{1}{N} \sum_{k=1}^N : (\phi_{N,k})^2 \phi_{N,j} : = \sqrt{2} \xi(\tau, x),$$

where $\xi(\tau, x)$ is space-(fictitious)-time white noise.

The Stochastic Damped Mean Field Wave Equation (SDMFW)

[Liu, 2025]: As $N \rightarrow \infty$, the interactions vanish and the target measures ρ_N converges to the non-interacting Gaussian measure $\mu_0 \otimes \mu_1$. The corresponding underdamped Langevin dynamics converges to:

$$\partial_\tau^2 \phi + \partial_\tau \phi + (1 - \Delta)\phi + \langle : \phi^2 : \rangle \phi = \sqrt{2} \xi(\tau, x).$$

Theorem 1 (Contribution 1: Unique Equilibrium of the SDMFW)

Let $\sigma \in (0, \frac{1}{2})$. The Gaussian measure $\mu_0 \otimes \mu_1$ is the unique equilibrium state of the SDMFW in the class of measures with a bounded normal-ordered second moment:

$$\langle \int_{\mathbb{T}^2} |\nabla^{-\sigma} : \phi^2 :|^2 dx \rangle < \infty.$$

Smoluchowski-Kramers Approximation (SKA)

Context [Zine, 2024; Liu, 2026]

The HL SM_N converges to the Parabolic $O(N)$ Linear Sigma Model (PL SM_N) (the transition from underdamped to overdamped Langevin dynamics). We establish the analogous result for their mean-field limits.

Theorem 2 (Contribution 2: From Wave to Heat: SDMF W $\rightarrow$ SMFH)

As the fictitious inertia $\varepsilon \rightarrow 0$, the damped wave equation (SDMF W) converges globally to the heat equation (SMFH). Concurrently, the fictitious momentum decouples, and converges to 0:

$$\varepsilon^2 \partial_\tau^2 \phi_\varepsilon + \partial_\tau \phi_\varepsilon + (1 - \Delta) \phi_\varepsilon + \langle : \phi_\varepsilon^2 : \rangle \phi_\varepsilon = \sqrt{2} \xi(\tau, x)$$

$$\Downarrow \quad \varepsilon \rightarrow 0$$

$$\partial_\tau \phi_0 + (1 - \Delta) \phi_0 + \langle : \phi_0^2 : \rangle \phi_0 = \sqrt{2} \xi(\tau, x)$$

$$d\mu_0 \otimes d\mu_\varepsilon$$

$$\varepsilon \rightarrow 0 \quad \Downarrow$$

$$d\mu_0$$

Summary:

The following diagram summarizes the relationships between the $O(N)$ models, their mean-field limits, and the respective SKA.

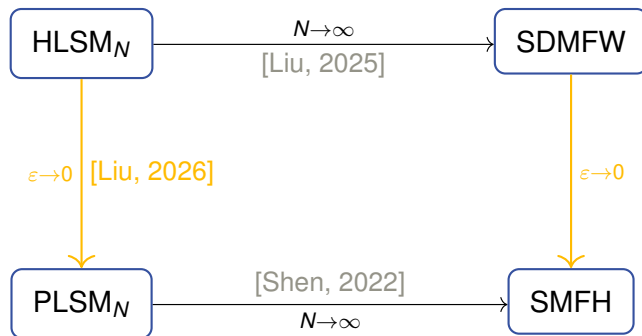


Figure: Diagram of limits between the Hyperbolic and Parabolic models.

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-  Tolomeo, L. (2023).
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